



National University
of computer and emerging sciences

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PROJECT REPORT

Project Name: Password Strength Analyzer & Generator

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2. Abstract

The "Password Strength Analyzer" is a desktop application designed to improve user security hygiene. It evaluates the complexity of passwords in real-time, detects vulnerabilities against common dictionary attacks, and generates strong, randomized passwords. The tool is built using Python to provide a secure, offline environment for password management.

3. Introduction

3.1 Problem Statement Users often use weak passwords like "123456" or personal names, making them easy targets for brute-force attacks. There is a lack of awareness

regarding what makes a password "cryptographically strong."

3.2 Project Objective

- To analyze password strength based on length, cases, digits, and symbols.
 - To warn users if their password is found in a list of common, leaked passwords.
 - To provide a one-click solution for generating secure passwords.
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4. Technical Details

4.1 Tools Used

- **Language: Python 3.x**
- **GUI Framework: Tkinter (for the interface)**
- **Modules: random, string (for logic), winsound (for alerts).**

4.2 How It Works (The Logic) The application uses a scoring algorithm (0 to 5 scale) to rate passwords:

1. **Positive Score: Points are awarded for length (>8 chars), uppercase, lowercase, numbers, and special characters.**
2. **Negative Score (Vulnerability Check): The system loads a `common_passwords.txt` file. If the user's password matches any entry in this file,**

the score is penalized, and a specific warning is shown.

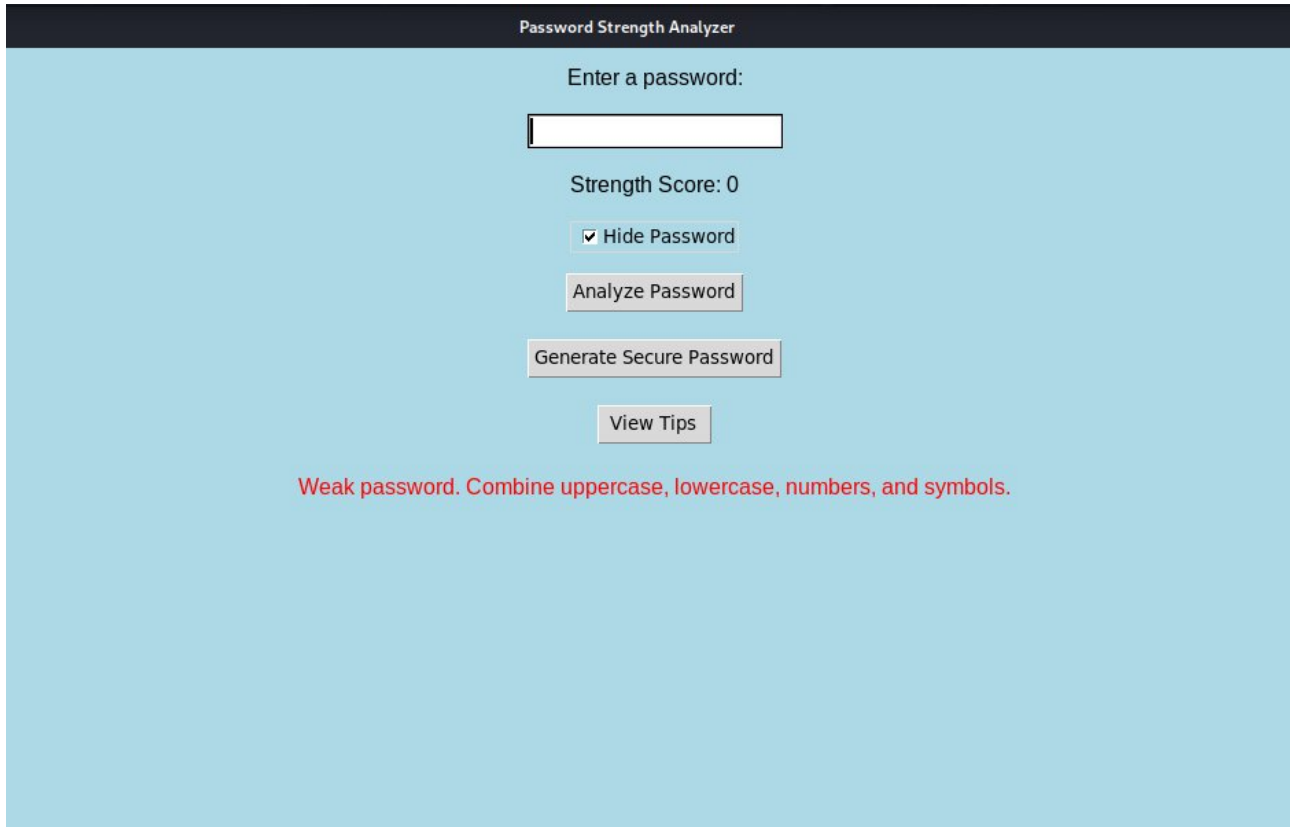
5. Implementation & Features

The project includes the following key modules:

- **Real-time Analysis:** As the user types, the `update_password_score` function runs immediately to give instant feedback (Red/Orange/Green text).
- **Password Generator:** The `create_password` function generates a 12-character string mixing letters, numbers, and punctuation.
- **Security Education:** A "View Tips" button opens a window listing guidelines for secure passwords, educating the user.
- **User Experience:** Includes a "Show/Hide" toggle

and a loading animation for a polished feel.

6. Results



The screenshot shows a web application titled "Password Strength Analyzer" with a dark header. The main content area is light blue and contains the following elements:

- A label "Enter a password:" above a white text input field.
- A label "Strength Score: 0" below the input field.
- A checkbox labeled "Hide Password" which is checked.
- A button labeled "Analyze Password".
- A button labeled "Generate Secure Password".
- A button labeled "View Tips".
- A red feedback message at the bottom: "Weak password. Combine uppercase, lowercase, numbers, and symbols."

- **Scenario 1: Inputting "password" shows "Weak, Common Password" (Red).**

Password Strength Analyzer

Enter a password:

rashid

Strength Score: 1

☒ Hide Password

Analyze Password

Generate Secure Password

View Tips

Weak password. Combine uppercase, lowercase, numbers, and symbols.

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- **Scenario 2: Inputting a complex string shows "Strong password!" (Green).**

The screenshot shows a web application titled "Password Strength Analyzer" with a dark header. The main area has a light blue background. It contains a text input field with the password "Rashid@Kh#n1". Below the input, it displays "Strength Score: 5". There is a checkbox labeled "Hide Password" which is checked. Below that is a button labeled "Analyze Password" with a dotted border. Further down is a button labeled "Generate Secure Password". At the bottom of the form area is a button labeled "View Tips". Below the buttons, a green message reads "Strong password! Excellent work!".

7. Conclusion

This project successfully addresses the issue of weak credentials. By combining an analyzer with a generator, it helps users create passwords that are resistant to standard dictionary attacks. It is a lightweight, offline tool ensuring that user data remains private.